



Investigation report

Rejected take-off and runway overrun, Tarlton Springs - AAIB-style
causal and contributory factors report

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INVESTIGATION WORKFLOW SUITE

Contents

Synopsis.....	3
Factual information	4
History of the flight.....	4
Recorded information.....	4
Aircraft information	5
Meteorology.....	5
Personnel information	5
Aerodrome information.....	6
Organisational and management information	6
Other information	6
Analysis	7
External Influences	7
Organisational Influences.....	8
Risk Controls.....	11
Local Conditions	12
Individual Actions	14
Occurrence Events.....	16
Conclusions	18
Findings	18
Causal factors	19
Contributory factors	19
Safety action and recommendations.....	21
Documented wildlife plan required only above a size threshold the aerodrome does not meet	21
Perimeter fence repair deferred for three budget cycles	21
Evening service approved as a schedule change, route risk assessment not reopened.....	22
Runway inspection required only at listed aerodromes, and the listing criterion is a strike.....	22
Rejected take-off training covers aircraft system failures below V1 only	22
Cabin floor proximity emergency lighting partially unserviceable	23
Distribution	23

Synopsis

On 18 June 2026 Saltbush Air SB512, a 50-seat twin-turboprop registered VH-SBQ, began its take-off from runway 09 at Tarlton Springs Aerodrome at 17:30 local, in darkness after last light, with 44 passengers, 2 flight crew and 1 cabin crew on board. Shortly after V1 the first officer called two kangaroos on the runway ahead, and the pilot in command rejected the take-off at 118 kt, 6 kt above V1. One second later the No. 2 propeller and right main landing gear struck both animals, and No. 2 reverse did not engage. The aircraft left the runway end at approximately 21 kt and stopped upright on the extended centreline, 62 m into the 90 m runway end safety area and 28 m short of the end of the prepared area.

There were no serious injuries. One cabin crew member reported bruising during disembarkation by the forward airstair, which was made without an evacuation. The aircraft was substantially damaged. Three No. 2 propeller blades were beyond limits, the engine required a shock-load inspection, the right main gear doors were distorted, both right main tyres deflated through the fuse plugs, and the lower right fuselage was punctured. Both kangaroos were killed and the aerodrome was closed for 4 hours 20 minutes.

The kangaroos on the sealed surface very likely caused both the rejected take-off and the strike. Rejecting 6 kt above V1 very likely began a stop for which the scheduled margin of 67 m was insufficient. The strike removed half the available reverse thrust two seconds into the stop, and deceleration was approximately 15% below the scheduled figure. Together with the late start, this shortfall carried the aircraft beyond the runway end with virtual certainty, although neither would have done so alone.

The northern perimeter fence had failed and its repair had been deferred for three budget cycles. The fence very likely gave the animals their route onto the runway. The departure, in the hour after last light when logged kangaroo movement peaked and nothing off the sealed surface could be seen, likely contributed. The operator required an after-last-light runway inspection only at aerodromes listed following a reported strike, so Tarlton Springs had never been listed, and this very likely contributed. No inspection was requested, an omission no document had prompted, and this likely contributed. The evening service was approved as a schedule change without reopening a route risk assessment restricting the aerodrome to daylight operations, which likely contributed. Rejected take-off training covering only aircraft system failures below V1 likely contributed. The pilot in command's account of a deliberate departure from the rule means this can also be read as a fleet matter. The standard's size threshold, under which the aerodrome needed no documented wildlife plan, likely contributed, although this rests on argument and contrary evidence was recorded.

A partially unserviceable floor proximity emergency lighting system and the crew's fifth sector 11 hours into duty did not contribute but were identified as risks. A tailwind and a No. 2 engine power loss before the reject were ruled out. Whether runway edge lighting intensity affected detection of the animals could not be determined.

The operator listed Tarlton Springs in Appendix 8A on 26 June 2026. The aerodrome operator installed a temporary electric wire on 25 June 2026 and programmed the fence replacement for 09/2026. It is recommended that the operator list aerodromes on wildlife exposure rather than a reported strike, and trigger a route risk assessment for any aerodrome first operated into after last light.

Factual information

History of the flight

On 18 June 2026 Saltbush Air SB512, a 50-seat twin-turboprop registered VH-SBQ, was the last scheduled sector of the day from Tarlton Springs Aerodrome to Sydney and the fifth sector of the flight crew's duty. The sector had been added that morning when the rostered crew reported unfit. On board were 44 passengers, 2 flight crew and 1 cabin crew. The doors closed at 07:16 UTC (17:16 local) and the service departed 40 minutes behind schedule, in darkness. The tower was open and rostered to close at 18:00 local.

The aerodrome operator's only runway inspection that day was at 21:35 UTC on 17 June (07:35 local on 18 June), before the first scheduled arrival, and recorded the movement area as clear. No further inspection was carried out or requested.

At 07:29:10 the tower cleared SB512 for take-off on runway 09, passing the wind as 090 at 6 kt. No runway inspection was offered or requested, and no wildlife transmission had been made on the tower channel in the preceding 90 minutes. The first officer was pilot flying. The aircraft weighed 20,800 kg and was configured with flap 10 for a dry runway, with a calculated V1 of 112 kt based on a 5 kt headwind component.

Power was set at 07:30:02. The start of the roll relative to the end of civil twilight is [NOT ESTABLISHED: time of last light and end of civil twilight at Tarlton Springs on 18/06/2026]. The 80 kt cross-check was completed at 07:30:15 and the pilot in command called V1 at 07:30:29. At 07:30:31 the first officer called two kangaroos on the runway ahead, moving from the northern side past the 1,100 m point towards the south, at an estimated range of around 200 m, possibly less. At 07:30:32 the pilot in command took control and rejected the take-off at 118 kt, six knots and three seconds after V1, at 1,152 m along runway 09 with 695 m of sealed surface remaining.

About one second later the aircraft struck both animals with the right main landing gear and the No. 2 propeller. Maximum braking was established at 07:30:34. The pilot in command selected reverse at 07:30:36, but only No. 1 reverse engaged. The aircraft left the end of runway 09 at approximately 21 kt at 07:30:51 and came to rest at 07:30:57, 62 m into the 90 m runway end safety area, upright on the extended centreline with the nose wheel in graded gravel, 28 m short of the end of the prepared area. The terrain beyond is ungraded.

There were no fire indications and no fire. The pilot in command set the parking brake, shut both engines down at 07:31:26, decided not to evacuate and instructed the cabin to remain seated. At 07:31:54 he asked the tower to activate the aerodrome emergency plan, which it did at 07:32:10, and aerodrome staff attended with a vehicle. No slides were deployed and the passengers disembarked by the forward airstair. The occurrence notification records disembarkation 4 minutes 8 seconds after the aircraft stopped, and the cockpit voice recorder captured the airstair operating at 07:34:40.

There were no serious injuries. One cabin crew member later reported bruising sustained during disembarkation. Both kangaroos were killed. The aerodrome was closed for 4 hours 20 minutes for inspection and removal of animal remains and propeller debris. Both flight crew were stood down and tested negative for drugs and alcohol. The operator's Head of Flying Operations suspended the evening Tarlton Springs service.

Recorded information

The FDR and CVR were removed by the operator's engineering representative at 12:40 UTC on 18 June 2026 and read out by the operator's flight data section.

From power set until 07:30:32 both engines were at take-off torque of 98.4% and 98.5%, with no exceedance, caution or warning, and fuel flow, ITT and propeller RPM within normal scatter. The headwind component was 4 to 6 kt throughout the roll, with no tailwind component, consistent with the 07:00 UTC routine observation.

The FDR showed the first officer's sighting call at 116 kt indicated airspeed, before any control input. The reject was initiated at 118 kt, with both power levers at flight idle within 0.6 seconds. At impact (07:30:33) there was a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g. Maximum braking was established 2 seconds after the decision, with brake pressure at 2,940 psi and anti-skid active on all four wheels. This was at the fast end of the operator's fleet distribution for rejected take-offs, against a fleet mean of 3.2 seconds.

No. 2 reverse was not selected and its interlock did not make. No. 2 propeller RPM decayed from 07:30:33. Mean deceleration from 07:30:34 to 07:30:44 was 2.6 m/s², approximately 15% below the certified accelerate-stop value for this weight and runway condition, which assumes symmetric reverse. Deceleration fell to 1.9 m/s² in the gravel. The stopping distance from 118 kt was 757 m.

The CVR captured the first officer's call, "Roos - roos, roos, on the left, they're on the -", and one second later the pilot in command's "Stopping. I have control." At 07:30:36 he said "Reverse - number two's not coming. Max brakes, max brakes." After the stop, the first officer said the animals "came out of the dark on the left of the strip, crossed the centreline", and that she had them in view for about two seconds or less. The pilot in command said "I never had them. I was heads-in on the engine page after V one."

Aircraft information

Engineering inspection between 19 and 21 June 2026 found:

- No. 2 propeller: three blades damaged beyond repairable limits, with leading edge deformation and one blade tip separation. Animal tissue was recovered from two blade roots.
- Engines: borescope inspection of both found no pre-existing damage and no evidence of a power anomaly before impact. The No. 2 engine was subject to a shock-load inspection.
- Right main landing gear: both tyres deflated through the fuse plugs, consistent with brake energy. The gear doors were distorted by the impact and the gear structure was within limits.
- Brakes: absorbed energy was calculated at 84% of the certified maximum, with no brake fire and no hydraulic leak.
- Lower right fuselage skin: abrasion and two punctures consistent with propeller debris.

The maintenance history of VH-SBQ is [NOT ESTABLISHED: maintenance history and deferred defects of VH-SBQ].

Meteorology

Runway 09 was dry and the aerodrome was in darkness.

Personnel information

The pilot in command held 8,410 hours, 2,140 on type, and had been in command for 3 years 1 month. Duty records give his last cyclic simulator as 11/03/2026 and last line check as 02/12/2025, both satisfactory, while his interview background gives his most recent check as a recurrent simulator check in February 2026. He had no prior occurrence history.

The first officer held 1,655 hours, 405 on type. Her last cyclic simulator was on 28/04/2026, all items satisfactory, and her interview background lists a line check in May 2026. Her two previous sectors into Tarlton Springs were in daylight, on 09/01/2026 and 17/02/2026. Duty records give her time with the company as 14 months and her interview background gives 17 months.

Duty records show a sign-on at 06:30 local for a five-sector duty planned to finish at 18:35 local, 12 hours 5 minutes against a permitted maximum of 12 hours 30 minutes. The occurrence took place 11 hours into the duty. Duty and rest in the preceding seven days were within company limits. The pilot in command's interview background gives a start time of 05:10 local. Both pilots described themselves as tired. The pilot in command said he would have declined a further sector but was not struggling.

Aerodrome information

The aerodrome kept a paper wildlife sighting register in its office. Between 01/2025 and 06/2026 it held 46 entries, 41 recording kangaroos on or adjacent to the movement area, of which 38 were timed between one hour before and two hours after last light. Eleven entries recorded animals on the sealed surface of runway 09/27. The log was not reported to any operator, and the aerodrome reporting officer confirmed no airline had asked for it.

The runway survey recovered the carcasses at 1,188 m and 1,221 m from the runway 09 threshold and found continuous tyre marks from 1,160 m to the runway end.

The pilot in command described the runway edge lighting as a corridor of light with darkness on either side and could not say whether this affected his seeing the animals. The first officer described it as normal, having only daylight landings there for comparison, and said she saw the animals clearly once they were on the sealed surface. The intensity setting in use is [NOT ESTABLISHED: runway edge lighting intensity setting at 07:30 UTC].

The aerodrome was below the size threshold for a documented wildlife plan. Repair of the perimeter fence had been deferred for three budget cycles.

Organisational and management information

Under Part A 8.14 the operator requested a runway inspection before the first departure after last light only at aerodromes listed in Appendix 8A, the listing criterion being a recorded strike. Tarlton Springs was not listed. The evening service had been approved as a schedule change without reopening the route risk assessment, and the pilot in command stated that no briefing about it reached him. The first officer had consulted the flight plan, charts, ERSA entry and company aerodrome brief page, none of which mentioned animals.

The operator's rejected take-off training covered aircraft system failures below V1 only. The pilot in command stated that every stop or go scenario he had been given in twenty years had been an engine failure below V1. He said he had seen kangaroos on the strip at Tarlton Springs four or five times, had mentioned this to ground staff, and had never filed a report.

Other information

During disembarkation the first officer noticed unlit sections of the aft floor proximity emergency lighting strip, with the emergency lighting armed and the ceiling lights on. A functional check was carried out on 24/06/2026, and its result is [NOT ESTABLISHED: outcome of the floor proximity lighting functional check].

Analysis

External Influences

Documented wildlife plan required only above a size threshold the aerodrome does not meet

Aerodrome Standard AS-139 places two separate obligations on the operator of a certified aerodrome in respect of wildlife.

- **Section 10.4.1** applies to every certified aerodrome. It requires the operator to take "reasonable measures to reduce the risk to aircraft operations presented by wildlife on or in the vicinity of the aerodrome."
- **Section 10.4.2** goes further and requires a documented wildlife hazard management plan. This applies only where the aerodrome "is used by aeroplanes having a maximum take-off weight exceeding 27,000 kg, or serves more than 350,000 passenger movements in a calendar year."
- **Section 10.4.3** covers aerodromes where neither limb of 10.4.2 applies. It provides that the measures "need not be documented", although the operator must be able to demonstrate what measures are in place.

Tarlton Springs fell below both limbs of 10.4.2. Its heaviest scheduled aeroplane was the 22,900 kg twin-turboprop used on the Saltbush Air service, and it handled 61,400 passenger movements in 2025. These figures were confirmed against the certified aerodrome register and the aerodrome operator's own compliance file, and they are not in dispute. It is virtually certain that the aerodrome was not required to hold a documented plan.

The aerodrome accordingly held no wildlife hazard management plan, no hazard assessment and no dispersal programme. Its only documented wildlife measure was the single daily inspection before the first arrival, under section 6.3 of the Aerodrome Manual. In this the aerodrome was not in breach of the standard. The matter is therefore not one of compliance but of the requirement itself. The threshold set by the standard left an aerodrome of this size with a wildlife programme that was undocumented, unaudited and, in practice, unfunded.

What the aerodrome already knew

The knowledge that a documented plan would have drawn on already existed at the aerodrome.

The aerodrome's paper sighting log for January 2025 to June 2026 contained 46 entries. Of these, 41 concerned kangaroos on or adjacent to the movement area. Of those 41, 38 were timed between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. The log was kept in the aerodrome office and was not reported to any operator. The aerodrome reporting officer confirmed that no airline had ever asked to see it.

The condition of the perimeter fence was also known.

- **Item WR-221**, raised in the council works register in 04/2023, recorded that the northern boundary fence between the 700 m and 1,300 m runway markers had failed and that animals had been observed passing through it. Replacement was estimated at \$41,000.
- The item was deferred as unfunded at the 2023, 2024 and 2025 budget rounds.
- **Item WR-263**, raised in 02/2026, recorded that the same section was deteriorating further.

- **The site inspection on 23/06/2026** found a continuous 140 m section of that fence with its bottom two wires separated from the posts and a clear animal pad through it at the 1,050 m mark.

The case for influence

The argument for influence is that a documented plan was the one mechanism that would have done two things:

- connected the aerodrome's own sighting log to the operators using the aerodrome, and
- placed the failed fence on a funded programme rather than leaving it on an unfunded works register.

A documented plan brings an audit trail and an owner. Each of the other controls examined in this occurrence failed for want of exactly those. The sighting log had no route out of the aerodrome office, and the fence had no programme to secure its repair.

The case against influence

The investigation also gave weight to the evidence against influence:

- Section 10.4.1 still obliged the aerodrome operator to take reasonable measures, whether or not those measures were documented.
- A documented plan is not the only way to repair a fence or to pass a sighting log to an airline.
- The council's three deferrals were budget decisions. There is no evidence that a documented plan would have been funded any better than the works register item was. A plan is a document, whereas a fence repair requires money.

How the evidence was weighed

The investigation concluded that it was likely the absence of a documented plan contributed to the occurrence. This conclusion rests on the balance of the evidence and is not a settled matter. It carries less confidence than the influence found for the two factors through which it acted:

- the deferral of the fence repair over three budget cycles, and
- the arrangement under which a runway inspection was required only at listed aerodromes, with a strike as the listing criterion.

Each of those was found very likely to have contributed. They are argued in their own subsections and are not argued again here.

The absence of a plan did not itself place the animals on the runway. Its effect ran through the fence deferral and the inspection arrangement. Because the evidence does not show that a documented plan would have secured funding for the fence, or would by itself have led to a runway inspection before the departure, its presence would not on its own have prevented the occurrence. The evidence therefore supports this factor as contributory and not causal.

The threshold in AS-139 10.4.2, which exempted Tarlton Springs from holding a documented wildlife hazard management plan, likely contributed to the occurrence as a contributory factor, acting through the deferred fence repair and the runway inspection arrangements, although the evidence that a documented plan would have changed either outcome is less firm than for the other contributing factors.

Organisational Influences

Perimeter fence repair deferred for three budget cycles

The External Influences analysis found that the aerodrome was not required to hold a documented wildlife plan. This subsection examines what happened to the one physical barrier between the surrounding land and the runway at Tarlton Springs once that requirement was absent.

The aerodrome operator's own works register shows that the condition of the northern perimeter fence was known. Item WR-221 was raised in 04/2023. It described four displaced panels and strained wire on the northern boundary between the 700 m and 1,300 m runway markers. It recorded that animals had been observed passing through, and it recommended replacement at an estimated \$41,000. The item was deferred at the 2023, 2024 and 2025 budget rounds, each time with the entry "defer, unfunded, review next cycle". In 02/2026, item WR-263 recorded the same section as deteriorating further. None of this had to be reconstructed. The fence's failure, the animals passing through it and the repeated deferral are all set down in the operator's own words.

The site inspection on 23/06/2026 found what the register predicted. A continuous 140 m section of the northern fence had its bottom two wires separated from the posts, and a clear animal pad ran through it at the 1,050 m mark (Figure 1). It is virtually certain that the fence was in this condition at the time of the occurrence.



Figure 1: The northern perimeter fence at the 1,050 m mark, photographed on the site visit: the fallen section and the animal pad through it. Site inspection photograph SV-2026-0417-07, northern boundary at the 1,050 m runway marker, taken 23/06/2026

The question is whether that breach was the route by which the two kangaroos reached the runway. Four pieces of evidence point the same way:

- The first officer saw the animals moving onto the runway from the northern side, past the 1,100 m point.
- The carcasses were recovered at 1,188 m and 1,221 m from the runway 09 threshold.

- The animal pad at 1,050 m lies on the same side of the runway and a short distance upstream of the strike position.
- A fence with its bottom wires detached from the posts does not exclude kangaroos.

The aerodrome's wildlife sighting log for 01/2025 to 06/2026 shows how often animals were present. It holds 46 entries:

- 41 record kangaroos on or adjacent to the movement area.
- 38 of those 41 fall between one hour before and two hours after last light.
- Eleven record animals on the sealed surface of runway 09/27.

The log is a paper register held in the aerodrome office. It is not passed to any operator, and no airline had ever asked for it. The exposure was therefore known locally and repeatedly observed, but it reached nobody who scheduled or flew services into the aerodrome.

On this evidence, it is very likely that the deferred repair allowed the animals onto the runway strip. The finding is not put higher than very likely, because the link between the breach and these two animals is inferred from position and from the animal pad rather than observed. The deferral was also not by itself sufficient to produce the overrun. The outcome additionally depended on the timing of the departure, the absence of a runway inspection, the decision to reject above V1, and the reduced deceleration after the strike. Each of those is argued in its own subsection. The deferral is therefore placed as contributory rather than causal.

The analysis supports the conclusion that the repeated deferral of the northern perimeter fence repair very likely allowed the kangaroos access to the runway, and was a contributory factor in the occurrence.

Evening service approved as a schedule change, route risk assessment not reopened

The current route risk assessment for Tarlton Springs is RRA-041 issue 3, dated 11/2022. It assessed runway length and accelerate-stop margin, the single runway with no lighting redundancy, the absence of an alternate within 200 nm at night, limited engineering support, and summer heat and density altitude. It carries the operating restriction "daylight operations only pending review of night lighting and diversion fuel policy". No issue 4 exists.

Change Management Record CMR-2025-118 introduced a scheduled evening rotation from 14/07/2025. This was the first company operation into Tarlton Springs after last light. The record states "Assessment: schedule change only" and "No new hazard identified". It answers "Route risk assessment review required: No." It was approved by the Head of Flying Operations on 22/05/2025.

CMR-2025-118 does not mention the daylight-only restriction anywhere. RRA-041 was not reopened, and the Safety Department's involvement went no further than receiving the approved record. All of these are current controlled documents and nothing contradicts them. It is therefore virtually certain that the change was approved in this way and that the route risk assessment was not reviewed.

The approver has since left the company and could not be interviewed. Why the change was assessed as a schedule change only, and what was considered at the time, could not be established. The analysis draws no conclusion about intent. What the documents do show is the structure of the process. It asked whether anything about the change was new, and it found nothing new about the aerodrome, the aircraft type, the crew qualifications or the engineering arrangements. It did not ask whether the existing assessment still held for an operation that the assessment itself prohibited.

Influence turns on what a review of RRA-041 would have found. A review against a night operation would have had to consider the aerodrome's exposure after last light. In the same pass it would have reached the wildlife exposure and the aerodrome's absence from Appendix 8A. Under section 8.14.2 of the Operations Manual, an Appendix 8A listing is what requires a runway inspection before the first departure after last light. The weaknesses of that listing criterion are argued under Risk Controls.

Two considerations weigh against a stronger finding. First, the restriction names night lighting and diversion fuel, and a review might have addressed those alone. Second, the approver cannot be asked what was before him. For these reasons the finding is put no higher than likely. The non-review was also not sufficient on its own to bring about the overrun, so it is placed as contributory.

The analysis supports the conclusion that approving the evening service without reopening RRA-041 likely removed the opportunity to identify the wildlife exposure after last light, and was a contributory factor in the occurrence.

Risk Controls

Runway inspection required only at listed aerodromes, and the listing criterion is a strike

The operator's Operations Manual Part A held a defence aimed directly at this kind of occurrence. Section 8.14.2 requires that "at an aerodrome listed in Appendix 8A, the pilot in command shall request a runway inspection from the aerodrome operator prior to the first departure after last light, and shall not commence the take-off roll until the result of that inspection has been passed." At any other aerodrome, 8.14.3 states that "no specific action is required." Appendix 8A admits an aerodrome only where "a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months." Nine aerodromes were listed. Tarlton Springs was not among them and had never been listed. The manual text and the appendix are documentary, and it is virtually certain that this arrangement existed.

The difficulty lay in how an aerodrome came to be listed, not in the inspection itself. No provision in Part A asked a crew to report a sighting, and none admitted an aerodrome to the list on any other evidence. The aerodrome's own paper register held 46 entries between January 2025 and June 2026. Forty-one of these concerned kangaroos on or adjacent to the movement area, 38 of those fell between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. The register was kept in the aerodrome office. It was not passed to any operator, and the aerodrome reporting officer confirmed that no airline had ever asked to see it. The local exposure was therefore known, but it had no route into the operator's system. Because the only trigger was a reported strike, the control could be armed only by the outcome it existed to prevent. The list was last reviewed on 03/03/2025, four months before the evening service began.

Had the requirement applied, the pilot in command would have requested an inspection and could not have begun the roll until its result was passed, and aerodrome staff disperse animals when they find them. The investigation allowed that an inspection completed some minutes before departure does not guarantee a clear runway at the moment of the roll. For that reason it is very likely, rather than certain, that the absence of this requirement contributed to the occurrence. The inspection could have reduced the exposure but could not have guaranteed its removal, so the factor is contributory rather than causal. Why no inspection was requested on the night belongs with the analysis of individual actions.

The listing criterion for the operator's runway inspection requirement meant the control could not apply at Tarlton Springs, and this very likely contributed to the occurrence.

Rejected take-off training covers aircraft system failures below V1 only

Module 4 of recurrent training syllabus TRG-SYL-07 contained four scripted simulator scenarios. These were an engine failure at 90 kt, an engine fire at 100 kt, a configuration warning at 85 kt, and an engine failure at V1 plus 5 kt with the take-off continued. All four are aircraft system events, and three arise below V1. In the five training cycles on file, no crew had been presented with an external obstruction scenario, and 214 crew completed the module in the 2025 to 2026 cycle alone. Part B 4.6 matched the syllabus. Section 4.6.4 states that "at or above V1 the take-off shall be continued," and the section contains no guidance on an obstruction seen on the runway at any speed. Both pilots stated independently that every stop or go scenario they had been given was an engine event arriving below V1. Because files beyond five cycles were not available, it is extremely likely, rather than virtually certain, that this gap existed.

Two readings of its influence were considered. On the first, the pilot in command had about two seconds after the first officer's call of "Roos - roos, roos, on the left" before calling "Stopping. I have control." He had no prepared frame for the situation, so the decision he made was the first time he had considered the question at all. On the second, the pilot in command stated that he knew the rule and departed from it deliberately because he judged continuing to be the greater hazard. On that reading, training on obstructions would not have changed what he knew, and the gap would be a finding about the fleet rather than a cause of this rejection. The investigation gave weight to the first reading, because a decision made unrehearsed in two seconds is not the same as one made against a briefed and practised frame. It did not set aside the second. This explains the circumstances of the decision and does not excuse it.

The absence of any external obstruction scenario in rejected take-off training and guidance likely contributed to the take-off being rejected above V1, and was a contributory rather than a causal factor.

Cabin floor proximity emergency lighting partially unserviceable

The first officer noticed sections of the aisle floor strip unlit while walking back through the cabin during the disembarkation. A functional check on 24/06/2026 found that two of the six floor proximity escape path segments on the left side aft of row 16 did not illuminate on battery power with the system armed. The fault was traced to a failed ballast common to both segments. It is virtually certain that the defect existed.

It is unlikely that the defect had any bearing on the occurrence or its outcome. No evacuation was commanded, the main cabin lights were on throughout, and passengers left by the forward airstair, without urgency, 4 minutes 8 seconds after the aircraft stopped.

The defect nonetheless matters. The pre-flight emergency lighting test on this type illuminates the ceiling lights and exit signs but neither illuminates nor monitors the floor segments, so neither the crew nor a line engineer could detect the fault without a dedicated functional test. The last such test was 11 months before the occurrence, within the 24-month programme interval. The same ballast is fitted across the type, so the condition of the fleet is unknown. In a fire, in smoke, or in a night evacuation over the wing, two dark segments in the aft cabin could separate an orderly exit from a blocked one.

The partial failure of the floor proximity lighting played no part in this occurrence, but it very likely increased risk as an undetectable gap in an escape path defence (Safety issue).

Local Conditions

Departure in the hour after last light, the peak of kangaroo movement

The take-off roll began at 17:30 local. The light and moon data for Tarlton Springs on 18/06/2026 give sunset at "16:58", the end of civil twilight at "17:25" and moonrise at "22:41 (waning crescent, 14% illuminated)". The roll therefore began 32 minutes after sunset and 5 minutes after the end of civil twilight, more than five hours before a thin moon would rise. The aerodrome was in full darkness, with no natural ambient light on the strip and no town lighting within sight. During the investigator's site visit at the same time of day on 23/06/2026, the unlit graded strip on either side of the sealed surface could not be seen from the centreline.

The aerodrome's wildlife sighting log, a paper register held in the aerodrome office, contains 41 entries for kangaroos on or near the movement area between 01/2025 and 06/2026. Of these, 38 fall between one hour before and two hours after last light, and eleven record animals on the sealed surface of runway 09/27. The completeness of a paper register cannot be verified. With that reservation, it is extremely likely that the departure took place within the aerodrome's period of peak kangaroo movement and in a light condition in which nothing off the sealed surface could be seen.

It is likely that this condition contributed. The animals were not seen until they were on the lit surface, and the cockpit voice recording captures the first officer's description shortly after the stop: "They came out of the dark on the left of the strip, crossed the centreline." On its own, a departure after last light is unremarkable. Its significance here is that it placed the aircraft on the runway when animals were most likely to be present and least likely to be seen, and this is why the absence of a runway inspection, discussed under Risk Controls, mattered.

The departure in the hour after last light was likely a contributory factor, because it made the animals' presence on the runway more likely and their detection less likely, although by itself it would not have prevented the occurrence.

Runway edge lighting intensity reduced the crew's ability to detect the animals

The investigation examined whether the runway edge lighting reduced the crew's ability to see the animals, and it could not settle the matter.

Some evidence supported the possibility. The commander recalled the lighting as brighter than he would have wanted. He described a corridor of light with everything outside it black, and said that anything off the seal could not be seen coming. A photometric check made at the investigator's request on 25/06/2026 found four northern edge lights between the 900 m and 1,200 m markers at between 118% and 131% of nominal intensity for stage 3. This was the section the animals crossed. The aerodrome operator's maintenance programme checks lamp serviceability but does not measure intensity. The deviation on 25/06/2026 is therefore established.

Other evidence did not support it. The first officer, who saw the animals, described the lighting as normal. She said she saw them clearly once they were on the sealed surface, at around 200 m. She also acknowledged that, having only two daylight sectors into Tarlton Springs to compare against, she was not well placed to judge whether the lighting was unusual. The commander's account of not seeing the animals has an explanation that does not depend on the lighting. He told the first officer, "I never had them. I was heads-in on the engine page after V one", which is where the procedure placed him.

The matter could not be resolved for three reasons. The only measurement was taken seven days after the occurrence, and nothing establishes the output of the lights on the night. The two crew accounts disagree. No study was available linking an edge light at 131% of nominal to the detection of an object on the runway surface. Resolving it would require evidence of the lights' output on the night and an established relationship between that intensity and detection. On the evidence

available, it is about as likely as not that the lighting reduced the crew's ability to detect the animals. The matter remains unresolved, and nothing from it is carried into the conclusions.

Take-off commenced with a tailwind above limits

A tailwind was ruled out, because the aerodrome observations at 06:30, 07:00 and 08:00 UTC and the wind recorded by the aircraft during the roll agree that the component on runway 09 was a headwind of between 4 and 7 kt throughout, consistent with the 5 kt headwind used in the crew's performance calculation.

Fifth sector, 11 hours into duty, 40 minutes behind schedule

The crew signed on at 06:30 local for five sectors. The fifth had been added that morning when the rostered crew reported unfit. The occurrence took place 11 hours 00 minutes into a duty planned to last 12 hours 05 minutes, against a permitted maximum of 12 hours 30 minutes, and the departure was 40 minutes behind schedule. In interview, both pilots described themselves as tired but not struggling.

The investigation considered whether fatigue played a part. The recorded data did not support any mechanism through which it could have done so. The power levers reached flight idle within 0.6 seconds of the decision to reject, and maximum braking was established within 2 seconds, against a fleet mean of 3.2 seconds. No step was omitted during the roll. The runway inspection was not requested, but no document prompted either pilot to consider requesting one, so the duty could not have affected that omission. It is unlikely that the duty contributed to the occurrence.

The exposure is nevertheless likely to be important. The evening Tarlton Springs rotation routinely places the last sector within the final hour of a 12-hour duty, at an aerodrome with no company engineering, no alternate within 200 nm at night and no crewing resilience. A single unfit crew member can therefore turn a four-sector day into a five-sector day on the morning of operation. The duty pattern did not contribute to this occurrence, but it is an other risk factor that warrants attention.

Individual Actions

No runway inspection requested before the first departure after last light

Neither the CVR nor the tower's recorded audio contains a request for a runway inspection, an offer of one, or any mention of wildlife before departure. The aerodrome's daily inspection log for 18/06/2026 shows a single inspection, at 07:35 local before the first arrival, with none afterwards and none requested. Two independent recordings and the aerodrome's own log agree, and it is virtually certain that no inspection was requested before SB512 departed.

An inspection was available. Aerodrome Manual 6.3 provides for inspections outside the scheduled time on the request of the reporting officer or a pilot in command, and the reporting officer confirmed that one would have been carried out if asked. However, nothing prompted either crew member to ask.

- Part A 8.14.2 requires a request only at aerodromes listed in Appendix 8A.
- For all other aerodromes, 8.14.3 states that "no specific action is required."
- Tarlton Springs had never been listed.

Both crew stated that they did not consider an inspection. The captain had expected Tarlton Springs to be on the list and was surprised to find that it was not. The aerodrome's wildlife sighting log recorded kangaroos on or adjacent to the movement area in 41 of its 46 entries, most of them

around last light. It was a paper register held in the aerodrome office, and no airline had asked to see it.

The omission therefore had no owner. No document placed the action before the crew, and the information that might have prompted it did not reach them. The reasons the operator's control was framed this way belong to the analysis of Risk Controls, where the listing criterion was found to be a safety issue.

An inspection would have covered the section from 1,050 m to 1,250 m, where the fence pad and both carcasses were found, and aerodrome staff disperse animals when they find them. It could not have guaranteed a clear runway at the moment of the roll, because an inspection some minutes earlier does not ensure the runway stays clear. On that basis it is likely that the absence of an inspection contributed to the animals being on the sealed surface when the aircraft reached them.

The absence of a runway inspection was a contributory factor. It made the presence of animals on the runway more likely, but an inspection alone would not with certainty have prevented the occurrence.

Take-off rejected 6 kt above V1

The FDR shows V1 at 112 kt at 07:30:29. Both power levers were retarded to flight idle from 07:30:32 at 118 kt, three seconds and six knots after V1. The CVR records the first officer's sighting call at 07:30:31 and the captain's "Stopping. I have control." at 07:30:32. The decision was therefore made and announced in a single exchange, about one second after the sighting. The captain's own account corroborates this, and it is virtually certain that the take-off was rejected above V1.

Part B 4.6.4 states that "At or above V1 the take-off shall be continued." Section 4.6 contains no guidance on an obstruction seen on the runway at any speed. It also contains no amplification for a case in which continuing is itself hazardous.

In interview and in his air safety report, the captain gave his reasoning. He judged that rotating into two large animals would leave the aircraft airborne at night with damage that could not be assessed, over country offering no usable option. He therefore chose the risk whose end he could see, and he acknowledged that the decision was outside procedure. It was taken in about one second, in conditions the procedure did not address. The rejected take-off training, which covered aircraft system failures below V1 only, is considered under Risk Controls. Whether the runway edge lighting affected how early the animals could be seen remains unresolved, as set out under Local Conditions, and nothing here depends on it.

The scheduled accelerate-stop distance of 1,780 m assumes the reject begins at or before V1, and the margin between ASDR and ASDA was 67 m. At the moment of the reject the aircraft was 1,152 m along runway 09, with 695 m of sealed surface remaining. The stopping distance achieved from 118 kt was 757 m. Beginning the stop six knots and three seconds late consumed that margin and more, before the loss of No. 2 reverse is counted. It is extremely likely that the late reject contributed to the overrun.

The rejection above V1 was a contributory factor. It followed from the animals' presence and from the scope of the training, and it acted together with the reduced deceleration to carry the aircraft beyond the runway end.

Maximum braking and No. 1 reverse applied without delay

The recorded data show the power levers at flight idle 0.6 seconds after the decision. Maximum braking was established at 07:30:34, two seconds after the decision, against the operator's recorded fleet mean of 3.2 seconds. Anti-skid was active on all four wheels, and brake energy reached 84% of the certified maximum without a brake fire. No. 1 reverse was selected at 07:30:36. No. 2 reverse

was unavailable because its propeller had been damaged in the strike. The first officer made the speed callouts and the runway-end call. It is virtually certain that the stop was executed in this way.

The aircraft left the sealed surface at 21 kt and stopped 62 m into the 90 m RESA, upright and on the extended centreline. Passengers were disembarked without an evacuation. A stop begun at the fleet mean response would, by estimate, have carried the aircraft a further 30 to 40 m, close to the far edge of the RESA. It is very likely that the speed of the response reduced the severity of the outcome.

The prompt application of braking and reverse was a positive factor. Following a reject outside procedure, the stop itself was executed faster than the fleet average and limited the overrun to a low-speed excursion onto a prepared surface.

Occurrence Events

Two kangaroos entered the runway strip and moved onto the sealed surface

It is virtually certain that two kangaroos were on the sealed surface of runway 09 during the take-off roll. Three lines of evidence each establish this independently:

- The carcasses of two adult kangaroos were recovered from the northern half of the runway, at 1,188 m and 1,221 m from the threshold.
- The CVR records the first officer's sighting call at 07:30:31, with the animals reported "on the left".
- The FDR records an impact at 07:30:33.

The first officer saw the animals crossing from the northern side towards the south and estimated their range at around 200 m at first sight. After the aircraft stopped, the first officer said they "came out of the dark on the left of the strip, crossed the centreline". The first officer added that they had been in view for "two seconds. Maybe less."

The presence of kangaroos on the runway at that hour was consistent with what the aerodrome already knew. Its wildlife sighting log for 01/2025 to 06/2026 holds 46 entries. Of these, 41 record kangaroos on or adjacent to the movement area, and 38 of those 41 fall between one hour before and two hours after last light. Eleven record animals on the sealed surface of runway 09/27.

The positions of the carcasses are consistent with the failed northern fence. The animal pad through that fence lies at the 1,050 m mark, 138 m before the first carcass and on the same side of the runway. The deferred fence repair, and the question of whether the animals came through that gap, are argued under Organisational Influences. The absence of any runway inspection after 07:35 local is argued under Individual Actions.

It is extremely likely that the animals' presence on the surface brought about both the rejected take-off and the strike, since each follows directly from it. Without the animals on the runway there would have been no reject and no strike. The factor is therefore causal.

The presence of two kangaroos on the sealed surface of runway 09 was a causal factor in the overrun.

No. 2 propeller and right main gear struck two kangaroos

It is virtually certain that the strike occurred. The recorded data, the physical damage and the recovered tissue all agree:

- The FDR records the impact at 07:30:33 at 114 kt, one second after the reject was initiated, with a lateral acceleration spike of 0.31 g right and a vertical spike of 1.48 g.

- Three No. 2 propeller blades were damaged beyond repairable limits, one blade tip had separated, and animal tissue was recovered from two blade roots.
- The right main gear doors were distorted, and the lower right fuselage skin bore abrasion and two punctures consistent with propeller debris.

The consequence for the stop was immediate. No. 2 propeller RPM decayed from 07:30:33 and the No. 2 reverse interlock did not make. When reverse was selected at 07:30:36, it was selected on No. 1 only. The commander's call at that moment, "number two's not coming", matches the recorded data.

The wheel brakes were not the source of the shortfall. Anti-skid was active on all four wheels, the system tested serviceable afterwards, and the brakes absorbed 84% of the certified maximum energy for the weight and speed. The scheduled accelerate-stop figures assume symmetric reverse. With one reverse unit lost within a second of the stop beginning, the achieved deceleration fell approximately 15% below the value used in the certified data.

It is virtually certain that the strike reduced the deceleration available. The reduction is direct and measurable, and it is the reason the aircraft left the runway rather than stopping on it. The factor is therefore causal.

The strike on the No. 2 propeller and right main gear removed No. 2 reverse thrust during the stop and was a causal factor in the overrun.

Deceleration achieved below the scheduled accelerate-stop performance

The scheduled accelerate-stop distance required was 1,780 m against an ASDA of 1,847 m, a margin of 67 m. That margin rested on two assumptions: that the stop would begin at or before V1, and that reverse would be symmetric. Neither held.

Distance reconstruction from recorded groundspeed places the aircraft 1,152 m along runway 09 when the reject was initiated, leaving 695 m of sealed surface. The stopping distance actually achieved from 118 kt was 757 m, which is 62 m more than the runway remaining.

The physical evidence agrees with that arithmetic. Continuous tyre marks run from 1,160 m to the runway end, and the nose wheel rut ends 62 m into the RESA on the extended centreline. The mean deceleration between 07:30:34 and 07:30:44 was 2.6 m/s². The aircraft left the sealed surface at 21 kt, and deceleration then fell to 1.9 m/s² in the graded gravel.

The shortfall had two inputs. The first was a stop that began 6 kt above V1, which is argued under Individual Actions. The second was deceleration approximately 15% below the scheduled value, which followed from the strike. The investigation assessed that both inputs were needed to carry the aircraft beyond the runway end, and that neither alone would have done so. The two are best understood as a single mechanism rather than as separate failures.

It is virtually certain that the shortfall existed and that it produced the overrun, since the shortfall is the overrun expressed as a performance quantity.

The deceleration achieved, below the scheduled accelerate-stop performance from a reject initiated beyond V1, was a causal factor in the overrun.

No. 2 engine power loss during the take-off roll

A No. 2 engine power loss during the take-off roll was very unlikely to have existed and was excluded. The FDR shows both engines at 98.4% and 98.5% torque, with no caution, warning or exceedance, until the power levers were retarded at 07:30:32. The borescope found no pre-existing damage, and the No. 2 RPM decay began only at the impact at 07:30:33, one second after the reject.

Conclusions

The investigation identified the following causal and contributory factors. They should not be read as apportioning blame or liability to any organisation or individual.

A finding stated without a likelihood term was established as fact.

Findings

1. Two of the six floor proximity escape path segments aft of row 16 on the left side of the cabin were dark on battery power with the system armed, because of a failed common ballast. The pre-flight emergency lighting test on the type neither illuminates nor monitors these segments, and the 24-month functional check interval cannot detect such a defect between checks. It is unlikely that the defect had any influence on the occurrence, because no evacuation was commanded and the main cabin lights were on throughout the 4 minute 8 second disembarkation. It is very likely nonetheless that it increased the risk to occupants in any evacuation made in darkness or smoke, and the same ballast is fitted across the type. (Safety issue)

2. It is very unlikely that the take-off was commenced with a tailwind above limits. The aerodrome observations gave a headwind component of 5 to 7 kt on runway 09, and the recorded aircraft data gave a headwind component of 4 to 6 kt throughout the roll, with no tailwind component at any point. This possibility was examined and ruled out.

3. The flight crew were operating the fifth sector of the day, which had been added that morning. They were 11 hours into a duty that was within company limits, and the flight was 40 minutes behind schedule. It is unlikely that this influenced the occurrence, because the recorded responses (0.6 seconds to flight idle and 2 seconds to maximum braking) were at the fast end of the fleet distribution, and because no document had prompted either crew member to consider requesting a runway inspection. It is likely that the evening Tarlton Springs rotation increased risk on that service as a standing matter, because it routinely placed the last sector in the final hour of a 12-hour duty, at an aerodrome with no company engineering and no alternate within 200 nm at night.

4. Maximum braking was established 2 seconds after the reject decision, against a fleet mean of 3.2 seconds, and No. 1 reverse was selected within 4 seconds. It is very likely that this reduced the severity of the outcome, since a stop begun at the fleet mean response would have carried the aircraft an estimated further 30 to 40 m, close to the far edge of the runway end safety area.

5. It is very unlikely that the No. 2 engine lost power during the take-off roll. Both engines were at 98.4% and 98.5% torque with no caution, warning or exceedance until the reject. The borescope inspection found no pre-existing damage. The No. 2 propeller damage and RPM decay began at 07:30:33, at the moment of impact and one second after the reject. This possibility was examined and ruled out.

The investigation could not determine either way whether the intensity of the runway edge lighting reduced the crew's ability to detect the animals.

A photometric check on 25/06/2026 found four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal for stage 3, in the section the animals crossed. The aerodrome's maintenance programme checked lamp serviceability but never measured intensity. The pilot in command recalled a corridor of light with darkness on either side of it.

Against this, the first officer described the lighting as normal and said she saw the animals clearly once they were on the sealed surface, at around 200 m. She also said that, with only two daylight landings at the aerodrome for comparison, she was not well placed to judge. The pilot in

command's not seeing the animals has an explanation that does not depend on the lighting, because after V1 he was looking at the engine page, which is where the procedure directs him.

The measured deviation was established, but its effect on detection was not. Resolving the matter would require evidence of the lighting's output on the night itself and an established relationship between edge light intensity and the detection of an object on the runway surface. Neither was available.

Causal factors

- Two kangaroos entered the runway strip and moved onto the sealed surface of runway 09, where they were sighted at an estimated 200 m and later recovered at 1,188 m and 1,221 m from the threshold, and it is very likely that their presence contributed to the overrun by prompting both the rejected take-off and the strike that followed.
- The take-off was rejected 6 kt above V1, at 118 kt and 1,152 m along the runway with 695 m of sealed surface remaining (the pilot in command having judged that continuing into two large animals at night carried the greater risk), when the scheduled accelerate-stop distance of 1,780 m assumed a reject at or before V1 and left a margin of 67 m, and it is very likely that the stop beginning at that point contributed to the overrun.
- The No. 2 propeller and right main gear struck two kangaroos at 07:30:33, one second after the reject began, destroying the No. 2 propeller and preventing No. 2 reverse from engaging, and with virtual certainty this contributed to the overrun by removing half of the available reverse thrust two seconds into a stop that had a 67 m scheduled margin.
- Deceleration achieved below the scheduled accelerate-stop performance, a mean of 2.6 m/s² from maximum braking to the runway end and approximately 15% below the certified value with reverse available on No. 1 only, meant that the stop from 118 kt required 757 m against the 695 m remaining, and this shortfall, produced by the late reject and the loss of No. 2 reverse acting together (neither alone would have carried the aircraft past the runway end), contributed with virtual certainty to the 62 m overrun.

Contributory factors

- The documented wildlife plan required under AS-139 10.4.2 applied only above a size threshold the aerodrome did not meet (its heaviest scheduled aeroplane was 22,900 kg and it handled 61,400 passenger movements in 2025), so the aerodrome was not in breach in holding no plan, and it is likely that this contributed by leaving the sighting log unconnected to operators and the fence repair without a funded programme, a conclusion reached on argument and against recorded contrary evidence that 10.4.1 still required reasonable measures and that nothing showed a documented plan would have been funded any better. (Safety issue)
- Perimeter fence repair was deferred for three budget cycles after works register item WR-221 recorded the northern fence as failed in 04/2023 with animals observed passing through, leaving a continuous 140 m section with an animal pad at the 1,050 m mark on the same side of the runway as both carcasses, and it is very likely that this provided the route by which the kangaroos reached the movement area. (Safety issue)
- It is very likely that the departure took place in the hour after last light, at the peak of kangaroo movement, the roll beginning at 17:30 local, 32 minutes after sunset and 5 minutes after the end of civil twilight with no moon, when 38 of the aerodrome's 41 logged kangaroo entries fell in that period, and it is likely that this contributed because nothing off the sealed surface could be seen and the animals were not sighted until they were on the lit surface.

- The evening service was approved under CMR-2025-118 as a schedule change, recorded as introducing no new hazard, and route risk assessment RRA-041 issue 3, which restricted Tarlton Springs to daylight operations pending review of night lighting and diversion fuel policy, was not reopened, and it is likely that this contributed because a review against a night operation was where the aerodrome's wildlife exposure and the Appendix 8A listing could have been reached, although such a review might have addressed lighting and fuel alone and the approver could not be interviewed to explain the assessment. (Safety issue)
- Operations Manual Part A 8.14 required a runway inspection before the first departure after last light only at aerodromes listed in Appendix 8A, the sole listing criterion being a reported wildlife strike involving a company aircraft in the preceding 24 months, so Tarlton Springs had never been listed despite eleven logged sightings of kangaroos on its sealed surface, and it is very likely that this contributed, since an inspection would otherwise have been required and aerodrome staff disperse animals when they find them, although an inspection some minutes before the roll could not guarantee a clear runway. (Safety issue)
- No runway inspection was requested before the first departure after last light, an omission that no operator document prompted either crew member to consider and that the aerodrome would have met if asked, and it is likely that this contributed because an inspection would have covered the section between 1,050 m and 1,250 m where the fence pad and both carcasses were found, although it could not have guaranteed that the runway remained clear at the roll.
- It is very likely that the operator's rejected take-off training covered aircraft system failures below V1 only, with no external obstruction scenario presented in the five training cycles available and no guidance in Part B 4.6 on an obstruction at any speed, and it is likely that this contributed by leaving the pilot in command to decide the question for the first time within two seconds, although he stated that he knew the rule and departed from it deliberately, so that the gap can also be read as a deficiency affecting the fleet rather than a cause of this reject. (Safety issue)

Safety action and recommendations

Documented wildlife plan required only above a size threshold the aerodrome does not meet

Aerodrome Standard AS-139 requires a documented wildlife hazard management plan only where an aerodrome is used by aeroplanes above 27,000 kg MTOW or serves more than 350,000 passenger movements a year. Tarlton Springs met neither condition. Its heaviest scheduled aeroplane was 22,900 kg and it handled 61,400 passenger movements in 2025. Under section 10.4.3 its wildlife measures therefore did not need to be documented at all, and the aerodrome was not in breach. The safety issue lies in the threshold itself. Section 10.4.1 still obliged the aerodrome operator to take reasonable measures, but the threshold meant those measures came without the documented plan, the identified owner and the audit trail that would otherwise accompany them.

The issue matters because the absence of a documented plan likely contributed to the occurrence. A plan was the mechanism that would have connected the aerodrome's wildlife sighting log to the operators using the aerodrome. Between January 2025 and June 2026 that log recorded 41 kangaroo entries. A plan was also the mechanism that would have placed the failed northern fence on a funded programme rather than an unfunded works register. This conclusion is held with less confidence than the other contributing factors in this investigation. The aerodrome operator could have repaired the fence or shared the log without a plan. The council's deferrals were budget decisions, and there was no evidence that a documented plan would have been funded any better.

No safety action has been recorded in response to this safety issue.

Perimeter fence repair deferred for three budget cycles

In 04/2023, the aerodrome operator's works register (item WR-221) recorded the northern boundary fence as failed between the 700 m and 1,300 m runway markers, with animals observed passing through it. The replacement was estimated at \$41,000 and was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. Item WR-263 recorded further deterioration in 02/2026. On the site visit, the section was a continuous 140 m failure with an animal pad through it at the 1,050 m mark (Figure 1). This was on the same side of the runway as the carcasses recovered at 1,188 m and 1,221 m.

The deferred repair very likely contributed to the occurrence. A fence in that condition did not exclude kangaroos, and it gave the two animals their route onto the runway strip and the sealed surface. The defect, and the fact that animals were using it, had been known to the aerodrome operator for more than three years without being remedied.

Safety action taken

The aerodrome operator funded item WR-221 from its 2026 contingency and programmed replacement of the 140 m northern boundary section for 09/2026. A temporary electric wire was installed on 25 June 2026 as an interim measure, and the replacement work was in progress.

This action answers the safety issue in part. It addresses the physical breach at Tarlton Springs. No action has been recorded that addresses the budgeting process under which the repair was deferred three times.

Safety recommendations

No safety recommendation has been made in respect of this safety issue.

Evening service approved as a schedule change, route risk assessment not reopened

The operator's Change Management Record CMR-2025-118 introduced a scheduled evening rotation into Tarlton Springs from 14 July 2025. The Head of Flying Operations approved it on 22 May 2025. This was the first company operation into the aerodrome after last light. The change was assessed as a schedule change only. No new hazard was identified, and the record stated that no review of the route risk assessment was required.

The current route risk assessment for Tarlton Springs, RRA-041 issue 3, carried the restriction "daylight operations only pending review of night lighting and diversion fuel policy". The change record did not reference that restriction, and RRA-041 was not reopened. No later issue exists.

It is virtually certain that the change was approved in this way, and it is likely that this contributed to the occurrence. Why the change was assessed as it was could not be established, because the approver has since left the operator and could not be interviewed.

The safety significance of the issue lies in the form of the change process. The process asked whether a change was new, rather than whether the existing assessment still held. In this case it placed a company aircraft into an aerodrome after last light for the first time, in direct contradiction of a live restriction in the operator's own current route risk assessment.

No safety action addressing this issue has been recorded. No safety recommendation is made in respect of it.

Runway inspection required only at listed aerodromes, and the listing criterion is a strike

The operator's Operations Manual Part A required the pilot in command to request a runway inspection before the first departure after last light at any aerodrome listed in Appendix 8A. It also required the pilot not to commence the take-off roll until the result was passed. At other aerodromes it required nothing. An aerodrome was listed only where a wildlife strike involving a company aircraft had been reported there in the preceding 24 months. No provision asked crews to report a sighting, and no other evidence could admit an aerodrome to the list.

Tarlton Springs had never been listed, although the aerodrome's own sighting log held eleven entries of kangaroos on the sealed surface. The operator's defence against this kind of occurrence could therefore be brought into effect only by the event it existed to prevent. The absence of that defence on the night very likely contributed to the occurrence.

On 26 June 2026 the operator amended Appendix 8A by temporary revision TR-2026-04 to list Tarlton Springs. As a result, the inspection requirement applies to every departure there after last light pending a permanent change. This action addresses the issue in part. It brings the control into effect at Tarlton Springs, but it leaves the listing criterion unchanged for every other aerodrome the operator serves.

Under the present criterion, no aerodrome can be listed before a strike has happened there. It is therefore recommended that the operator replace the Appendix 8A listing criterion so that aerodromes are listed on their wildlife exposure rather than on a reported strike. It is also recommended that the operator ensure a route risk assessment is triggered for any aerodrome first operated into after last light.

Rejected take-off training covers aircraft system failures below V1 only

The operator's recurrent training and its Operations Manual prepared crews for the rejected take-off decision only in terms of aircraft system events. Module 4 of TRG-SYL-07 contained four scripted scenarios: engine failure at 90 kt, engine fire at 100 kt, configuration warning at 85 kt, and engine failure at V1 plus 5 kt with the take-off continued. All four were aircraft system events, and three of them arose below V1. No crew had been presented with an external obstruction in the five training cycles on file, which in the current cycle alone covered 214 crew. Part B section 4.6 required the take-off to be continued at or above V1 and gave no guidance on an obstruction on the runway at any speed. It is extremely likely that this gap existed.

The gap likely contributed to the take-off being rejected 6 kt above V1. The captain made the decision without a briefed or practised frame to draw on. He also stated that he knew the rule and departed from it deliberately because he judged the alternative worse. On that reading, the gap concerns the preparation of the fleet rather than the cause of this reject. Both readings are retained. The gap bears on the conditions in which the decision was made, and it does not change the procedure against which that decision is measured. Its significance extends beyond this occurrence, because the same syllabus prepares every crew in the fleet.

No safety action has been recorded in response to this safety issue, and no recommendation is made in respect of it.

Cabin floor proximity emergency lighting partially unserviceable

During the disembarkation, the first officer noticed while walking back through the cabin that sections of the aisle floor strip were unlit. A functional check of the cabin emergency lighting on VH-SBQ on 24 June 2026 found that two of the six floor proximity escape path segments aft of row 16 on the left side were dark on battery power with the system armed. The fault was traced to a failed common ballast. It is virtually certain that the condition existed.

The pre-flight emergency lighting test on this type illuminates the ceiling lights and exit signs. It neither illuminates nor monitors the floor segments. The defect could therefore not be detected by the crew, or by a line engineer, without a dedicated functional test. The last such test had been carried out 11 months before the occurrence, within the 24-month programme interval.

The defect played no part in the overrun or its outcome. No evacuation was commanded, and the disembarkation was made by the forward airstair with the main cabin lights on. It took 4 minutes 8 seconds and was completed without incident. The defect is nonetheless very likely to be important to future safety. In a fire, in smoke, or in a night evacuation over the wing, two dark segments in the aft cabin could separate an orderly exit from a blocked one. A 24-month interval on a system that the daily test cannot see means that the true state of the fleet is unknown. The same ballast is fitted across the type. [NOT ESTABLISHED: condition of the floor proximity lighting ballast across the remainder of the fleet]

No safety action has been recorded in response to this issue, and no safety recommendation is made.

Distribution

Title	Name
Head of Flying Operations	R. Calvert
Head of Safety	M. Falkiner
Fleet Manager, Turboprop	D. Osei
Training Manager	S. Arnott
Aerodrome Reporting Officer, Tarlton Springs	K. Pryor